

“Can we get married?”

Pre-marital, pre-pregnancy and prenatal counselling

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贊譽百載
育護未來

“Should we have children?”

- Having a safe and healthy pregnancy, from conception to birth
- ↑chance of having a healthy child
- ↓chance of having adverse pregnancy outcome
- Understand available prenatal screening and diagnostic tests



Objective of pre-marital and pre-pregnancy counselling

- Detect and assess any specific health problems in the woman or her partner that may be relevant → managed prior to the pregnancy
- Obtaining general advice about optimizing personal health care and lifestyle



Timing of pre-pregnancy counselling

- appropriate whether the **reproductive-aged** patient is currently using contraception or planning pregnancy.
- as health status and risk factors can change over time, pre-pregnancy counseling should occur several times during a woman's reproductive lifespan



Outline

1. Reproductive risks – maternal, fetal
2. Risk identification
3. Options for at-risk couples
4. Counselling concerns



Reproductive risks - maternal

- Pre-existing medical conditions
- Pregnancy related complications
- Operative risk
- Psychosocial problems



Reproductive risks - fetal

- Growth problems
- Congenital anomalies
- Infection
- Birth trauma



- Fetal growth problems

Problem	Maternal cause	Fetal cause
Growth restriction	Hypertension (early onset)	Syndromal/infection
Macrosomia	Gestational diabetes (suboptimal control)	Syndromal

- Congenital anomalies
 - Chromosomal/genetic
 - Structural
 - Functional



Risk identification

- Detailed history taking
- Physical examination
- Screening tests

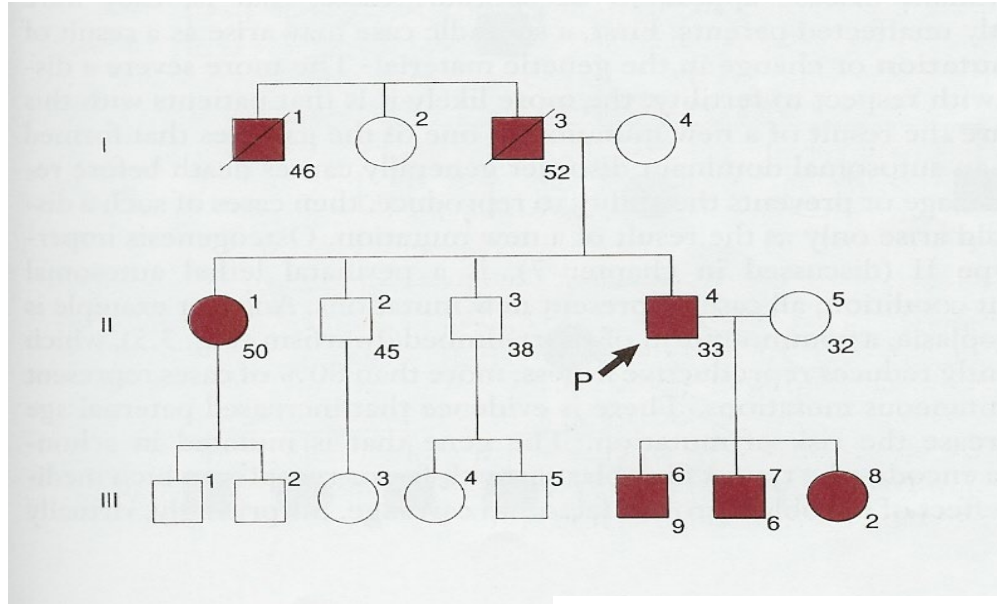


History

- Age
- Ethnicity
- Consanguinity
- Family history of congenital anomalies/genetic disorders
 - If high risk of a chromosomal or genetic disorder based on family history, referral to **pre-pregnancy genetic counselling** is preferred to allow more reproductive options

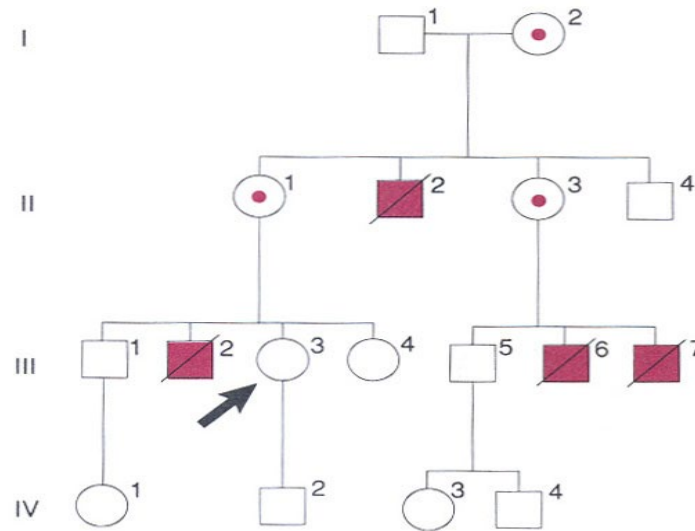
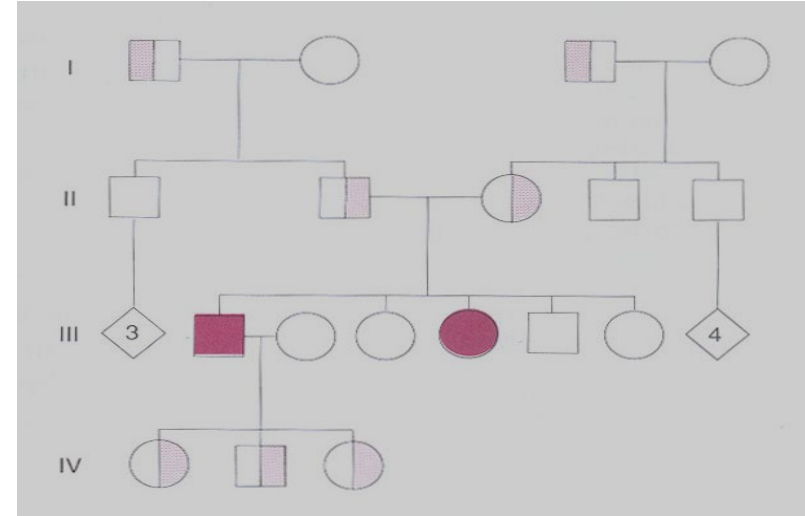


Autosomal dominant inheritance



Principles of Medical Genetics,
2nd ed. 1998

Autosomal recessive inheritance



X-Linked
inheritance



History

- Medical history
 - Discussion on how the medical condition may affect, or be affected by a pregnancy
 - Stabilisation of pre-existing medical condition
 - Multidisciplinary pre-pregnancy planning



History

- Medication use
 - Review current medications including over the counter on appropriateness and teratogenic potential
 - May need adjustment of medication prior to pregnancy to optimize disease control and minimize teratogenic risk



History

- Surgical history
- Drug allergy
- Occupation



History

- Lifestyle, risk of STI, intimate partner violence
- Smoking, alcohol and substance use
 - Should be stopped before conception
 - No known safe level of alcohol consumption during pregnancy
 - Considered for both parents when relevant
- Nutrition
 - Well balanced diet
 - Weight management
 - Periconceptional folic acid supplementation



History

- Vaccinations
 - Hepatitis B, rubella and varicella immunization can be considered for women with incomplete immunity
 - Seasonal influenza vaccination in pregnancy is safe
- Travel and environmental risks
 - Reduce the chance of infection



History

- Obstetric history
 - Congenital anomalies
 - Stillbirth, neonatal death
 - Recurrent pregnancy loss
 - Grand multiparity
 - Difficulty delivery, birth trauma



Physical examination

- General examination
- Auscultation of heart sound
- Breast examination
- Cervical screening
- +/- other tests as indicated

- Blood pressure, body mass index



Screening tests

- Complete blood picture (including Hb, MCV)
- Blood group, Rhesus factor
- Infection – VDRL, Hepatitis B, rubella antibody, HIV antibody
- Cervical smear
- Urine protein, sugar



Screening tests

- Available screening tests for low risk women for carrier of more common genetic conditions (e.g. thalassaemia)
- Expanded carrier screening
 - currently not funded by public health system
 - cost effectiveness, acceptance and uptake uncertain
 - Detailed pretest counselling on benefits and limitations of testing





Antenatal screening tests in Hong Kong

Gestation	Tests
First booking	Hb, MCV, Hep B, Rubella, VDRL/FTA-ABS, Rhesus blood group, red cell antibody, HIV (opt out)
11 to 13 ⁺⁶ weeks	1 st tier Down syndrome screening: 1 st trimester combined DS screening OR
16 to 19 ⁺⁶ weeks	2 nd trimester biochemical Down syndrome screening
18 to 22 ⁺⁶ weeks	Morphology scan
26 to 30 weeks	Oral glucose tolerance test for gestational diabetes
35-37 weeks	Group B Streptococcus screening



準媽媽產前診斷新流程

Process of new prenatal diagnosis for mums-to-be

所有孕婦 All pregnant women

第一層篩查：在第11至13周
照超聲波量度胎兒頸皮、檢驗媽媽血清標誌物
First-tier screening by ultrasound examination of foetal nuchal translucency which measures skin thickness at the back of a baby's neck; and maternal serum markers in the 11th-13th week

→ 高風險 High risk

→ 低風險 Low risk

Or biochemical screening when
16-19 weeks if booked late

NEW! 第二層篩查：無創性胎兒染色體篩查、
檢測媽媽血漿內的胎兒基因

Second-tier screening by non-invasive prenatal
testing (NIPT): test maternal plasma foetal DNA

- 抽絨毛膜或羊膜穿刺（俗稱抽羊水）作以下測試
 - PCR 唐氏綜合症快速測試
 - 傳統染色體核型分析：分析染色體缺失、易位及遺傳病

NEW! - 胎兒基因晶片測試：診斷基因序列有否缺失和重複

- 胎兒結構超聲波
- Chorionic villus sampling or amniocentesis
 - PCR: rapid aneuploidy test for Down syndrome
 - Karyotyping: to detect aneuploidies, gross structural chromosomal abnormalities and translocations
- NEW!** - Chromosomal microarray (CMA): molecular karyotyping to detect microdeletions and microduplications

- Foetal anomaly ultrasound examination

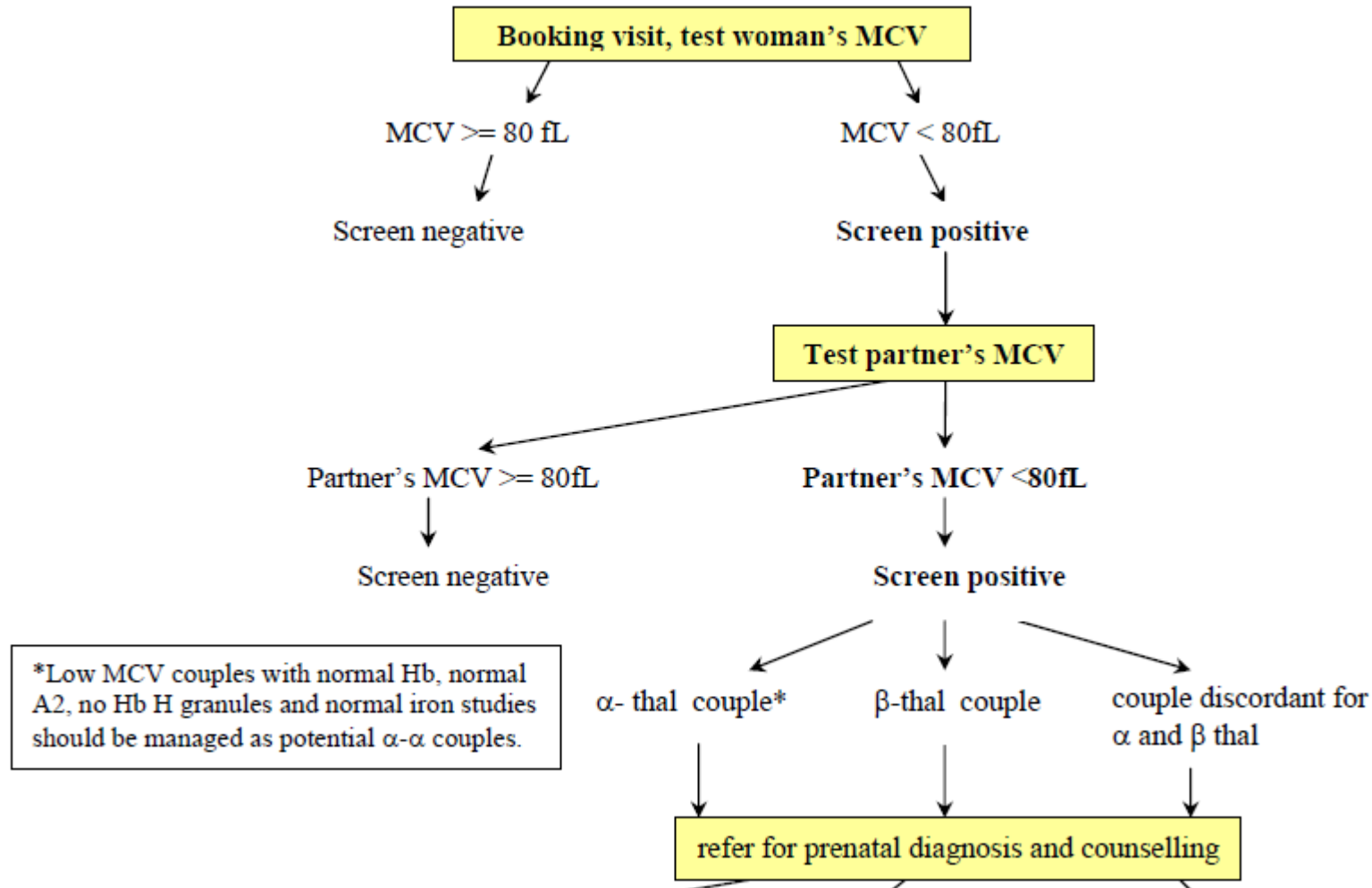
一般會為35歲或以上的孕婦在
第18至22周進行胎兒結構超聲波
Foetal anomaly ultrasound
examination offered to pregnant
women aged 35 or above in
the 18th - 22nd week

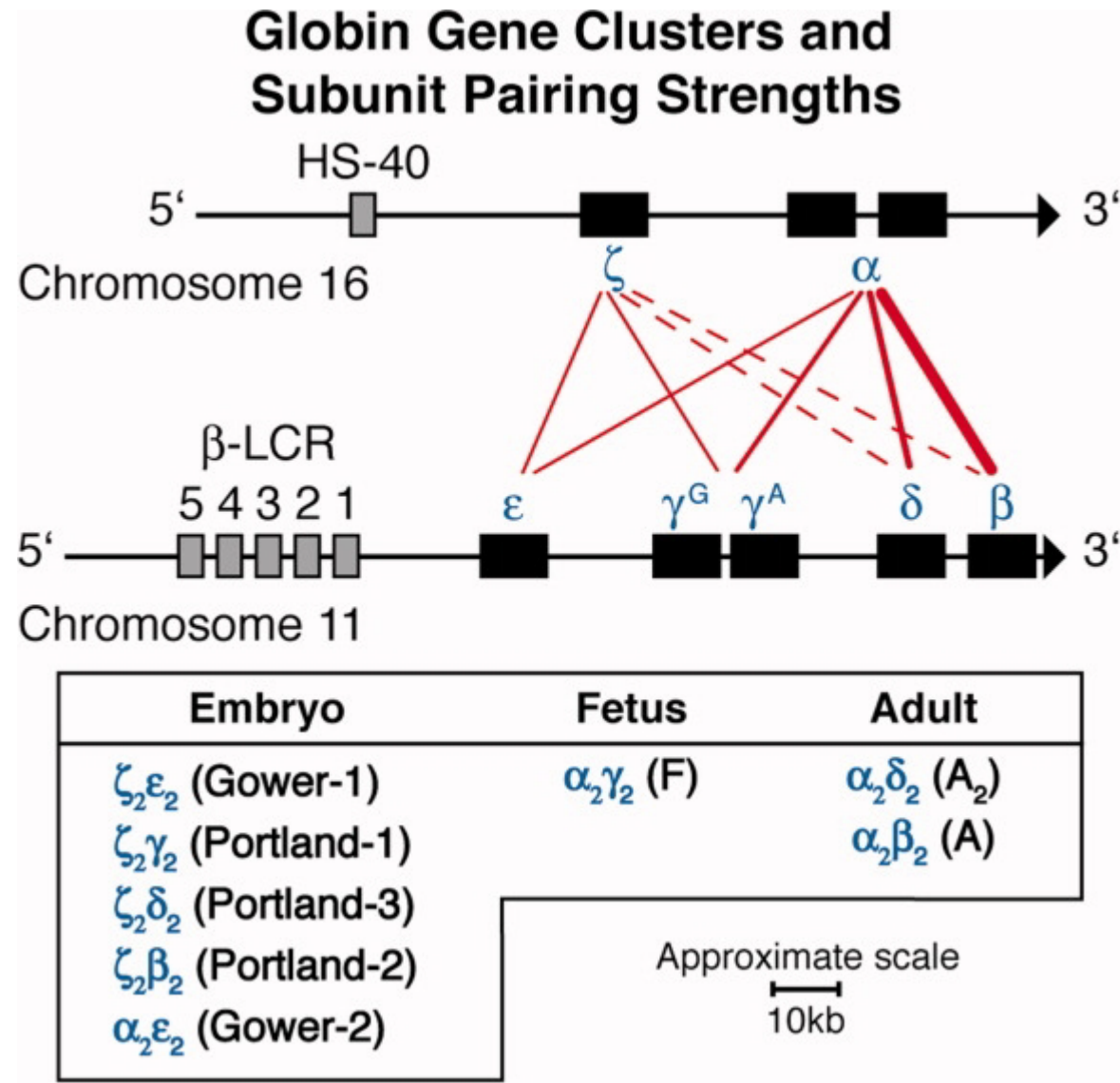
常規產檢
Routine
antenatal care

如有需要，轉介至遺傳輔導專家跟進
Refer to genetic counselling if necessary

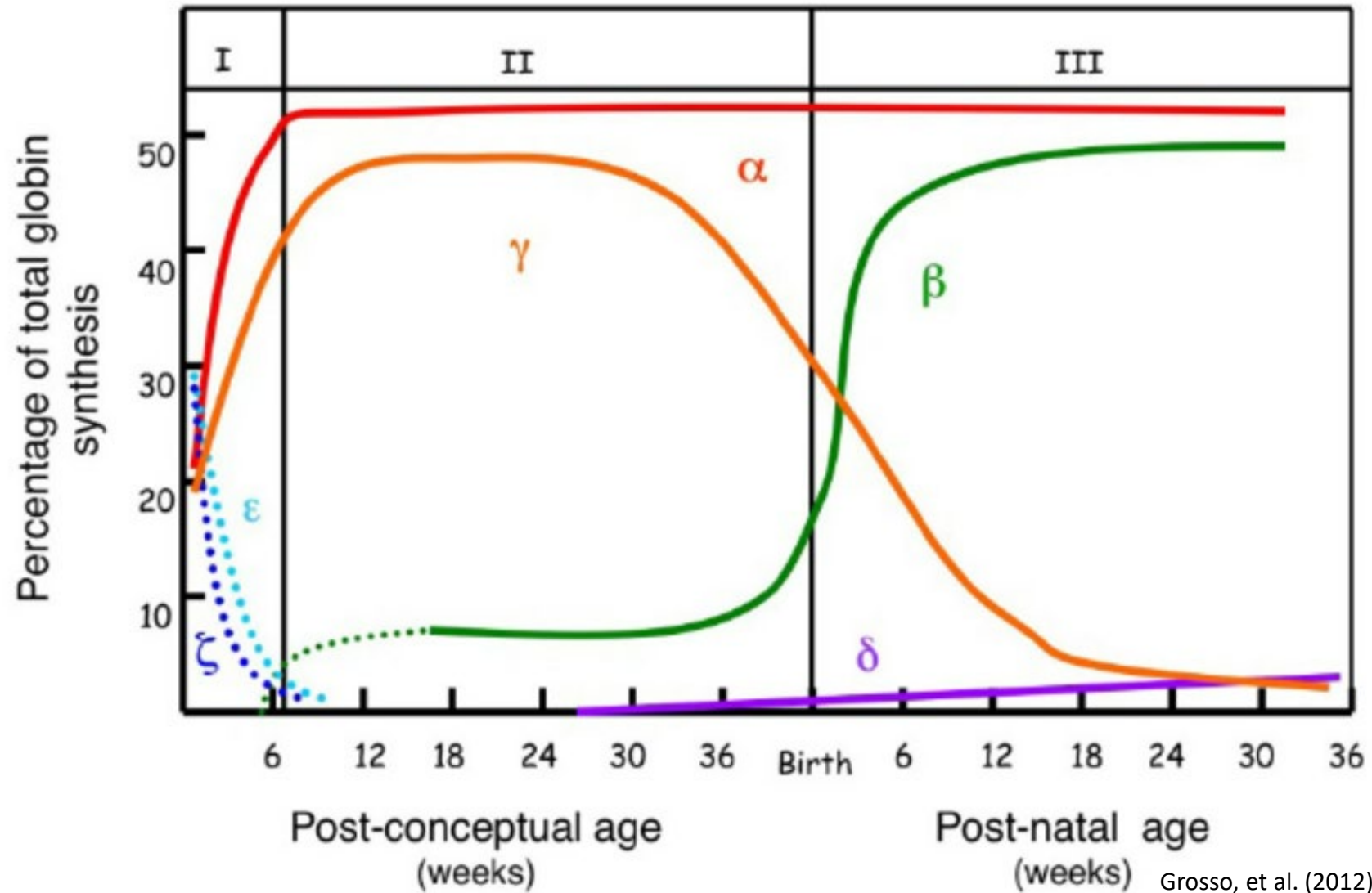


Guidelines of Antenatal Thalassaemia Screening



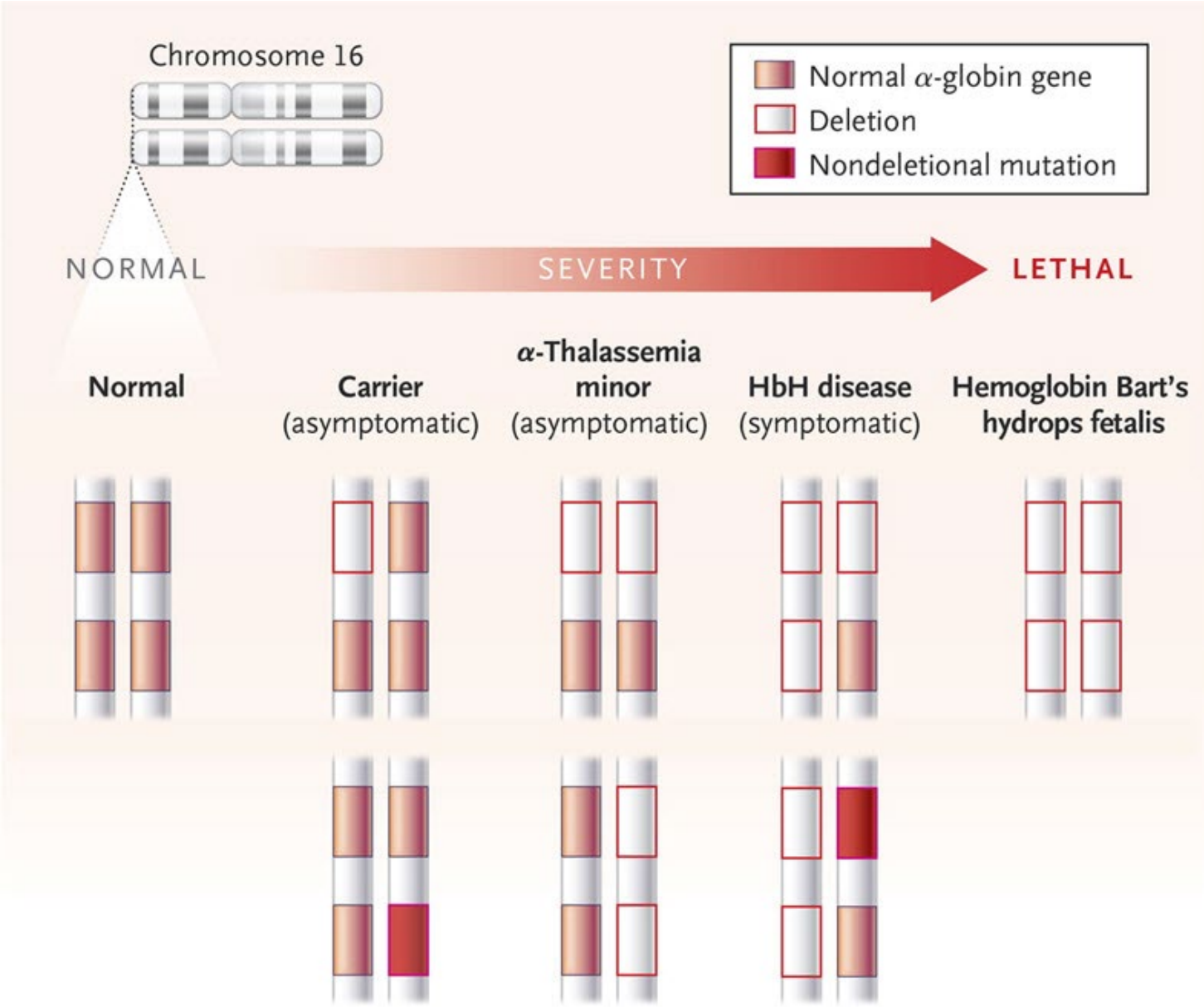


Globin gene expression





Phenotype–Genotype Relationship in α -Thalassemia



Diagnosis of α -thalassaemia carrier

- HbA $\alpha_2\beta_2$
- HbA₂ (normal) $\alpha_2\delta_2$
- HbF (normal) $\alpha_2\gamma_2$
- HbH granules (+) β_4
- IC strip positive
- MCV <82 fL
- Fe normal or raised



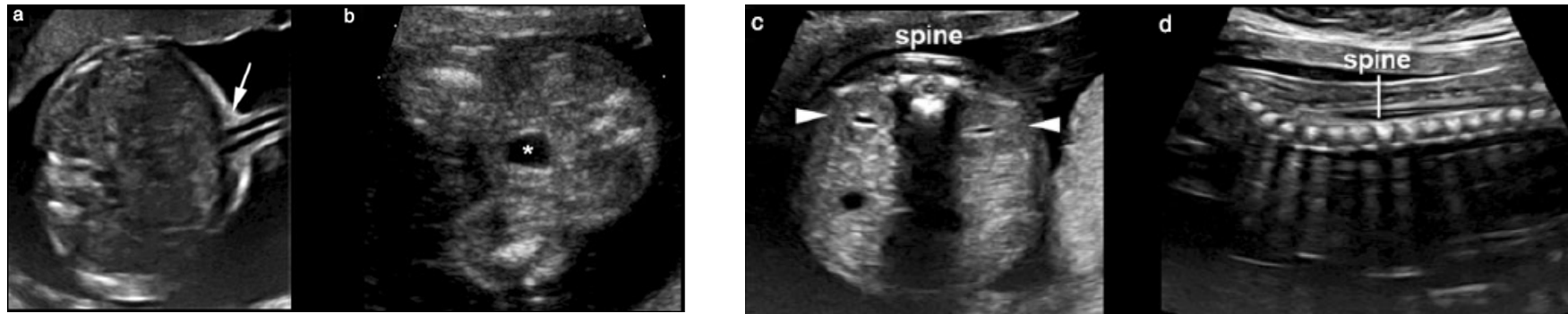
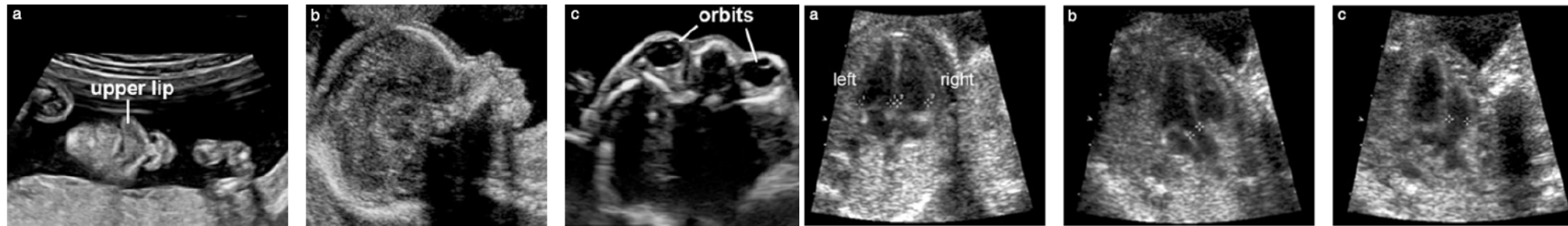
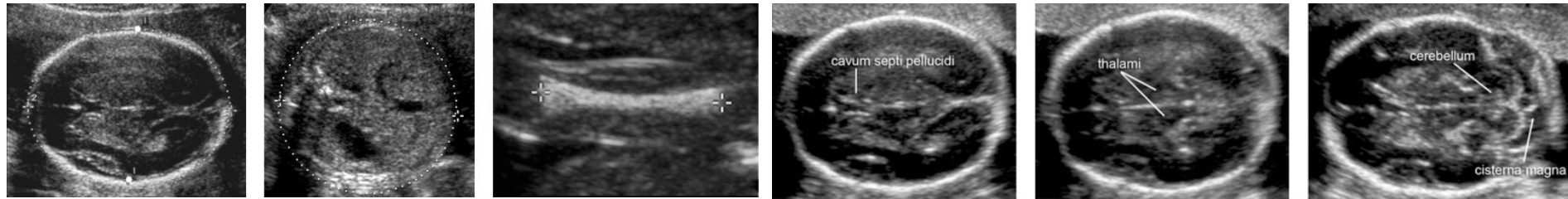
Diagnosis of β -thalassaemia carrier

- HbA $\alpha 2\beta 2$
- HbA₂ (raised) $\alpha 2\delta 2$
- HbF (raised) $\alpha 2\gamma 2$

- MCV $<82 \text{ fL}$
- Fe normal or raised



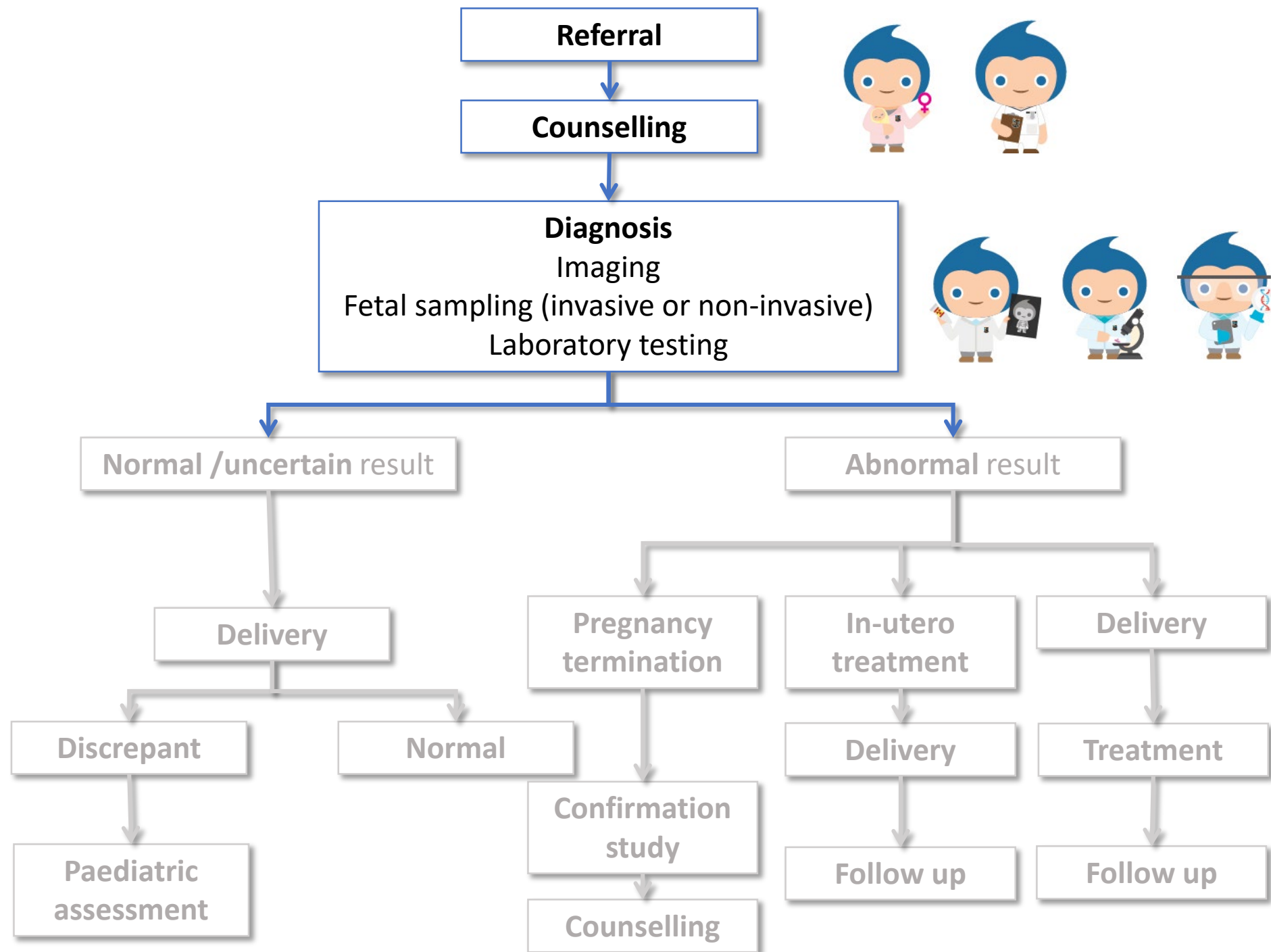
Fetal morphology scan 18-22 weeks



Prenatal diagnosis

- The objective of **diagnosing** disorders is to **help affected individuals** by offering timely **treatment**, to prevent irreversible damage, or **support**, if treatment is unavailable





Invasive tests

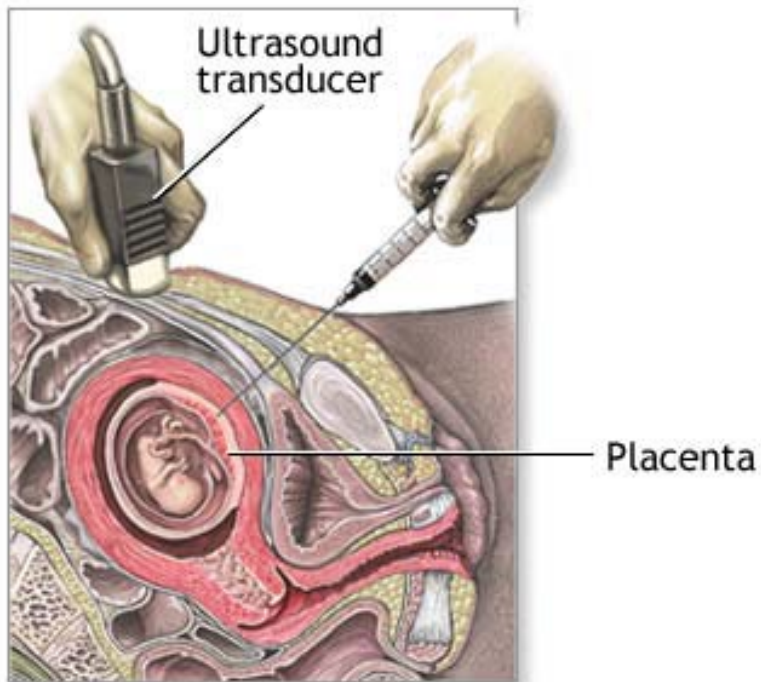
- Chorionic villus sampling
 - Amniocentesis
 - Cordocentesis
-
- Chromosomal test +/- molecular genetic test as indicated



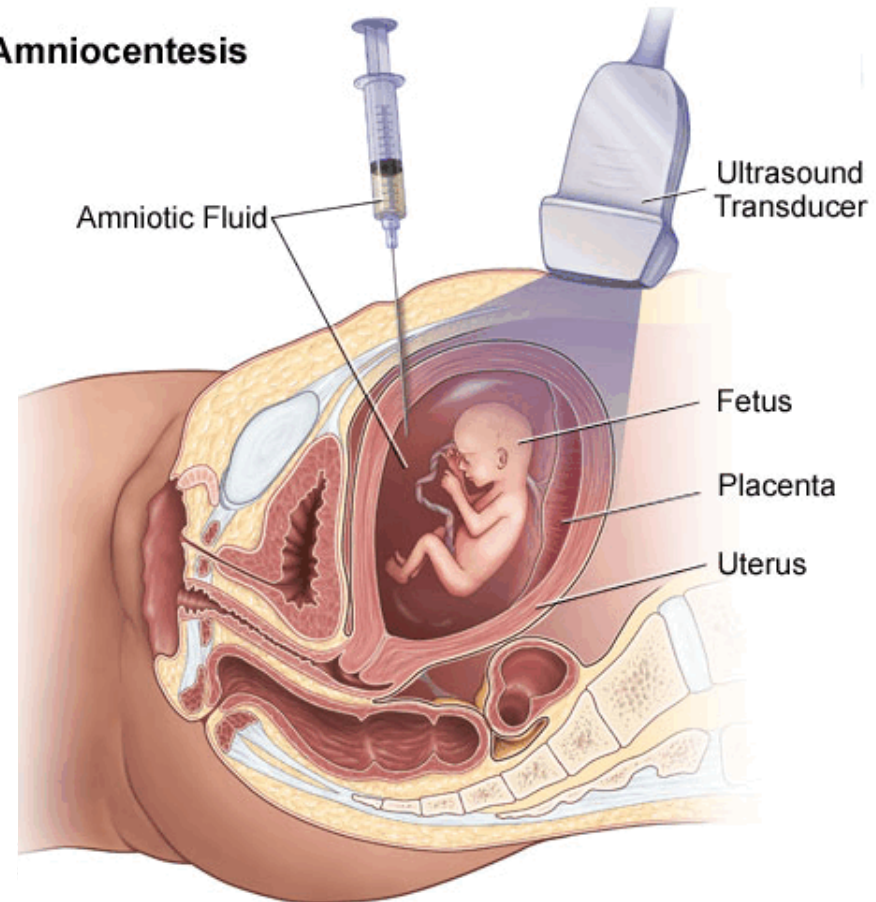
Risks of the having the tests

Chorionic villus sampling (CVS)

Transabdominal procedure



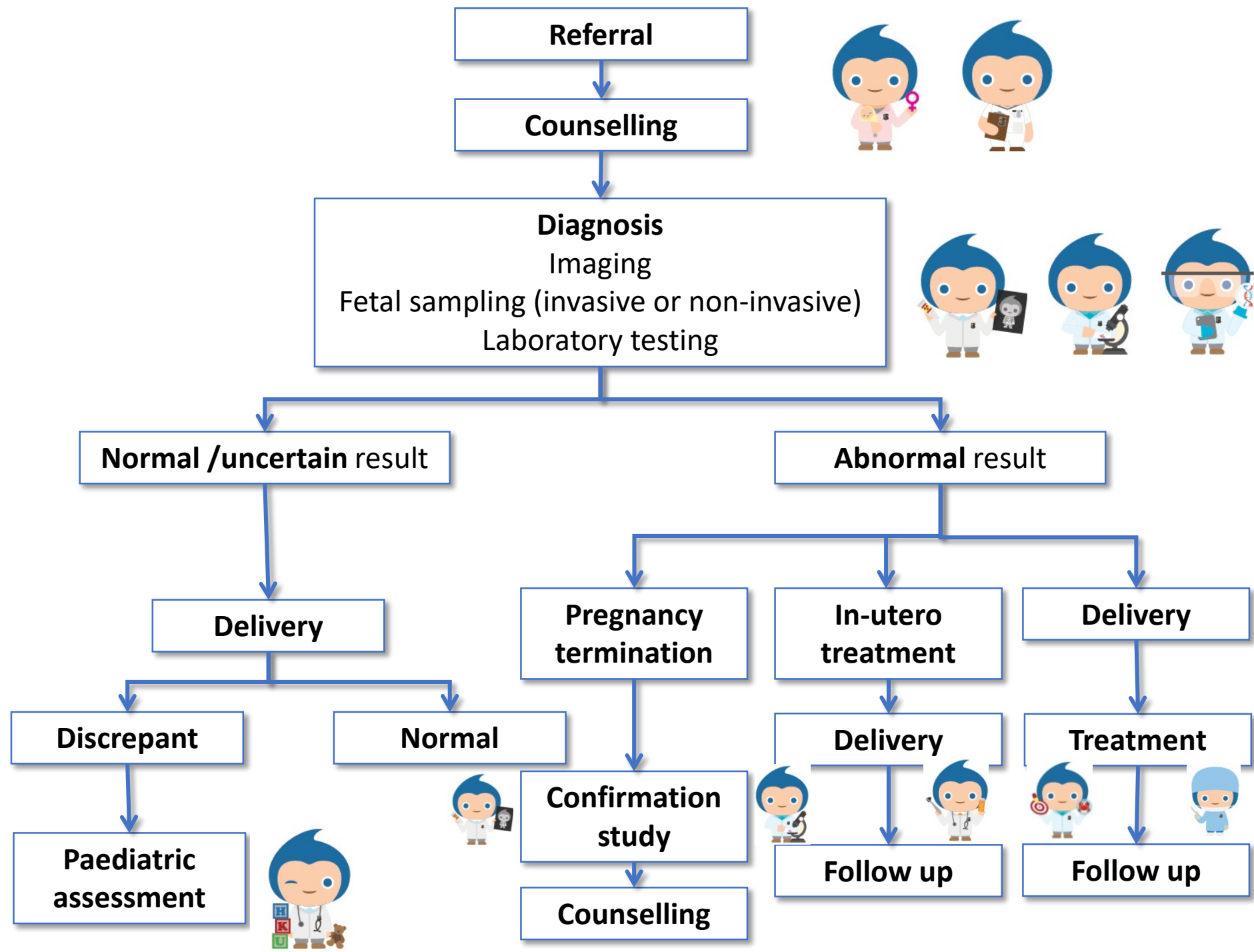
Amniocentesis



Comparison between CVS and amniocentesis

	CVS	Amniocentesis
Procedure time	11 to 14 weeks	16 to 20 weeks
Samples and laboratory test	Chorionic villi for cytogenetic studies	Amniocytes for cytogenetic studies
Pain relief	Local analgesia	Not required
Miscarriage risk	0.1-0.2 %	0.1-0.2%
Culture failure rate	Less than 1%	Less than 1%
Normal report interpretation	A normal report does not guarantee that the fetus is free from disorders that are not chromosomal in origin	
Test accuracy	✧ Both tests are very accurate ✧ A small proportion of test results may show mosaicism which may require further investigation	
Turnaround time for CMA	7 working days	
Maternal risk	Minimal	





Counselling concerns

- Counsel – explain, enable others to decide
- Advise – explain, recommend a course of action



Improving the chance of having a healthy child

- Prepare **before** conception
- Screen or test **after** conception
- Current tests **cannot guarantee** a healthy baby for every pregnancy



Essence of counselling

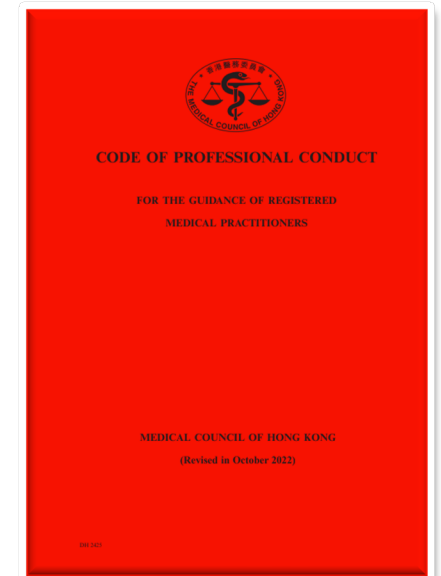
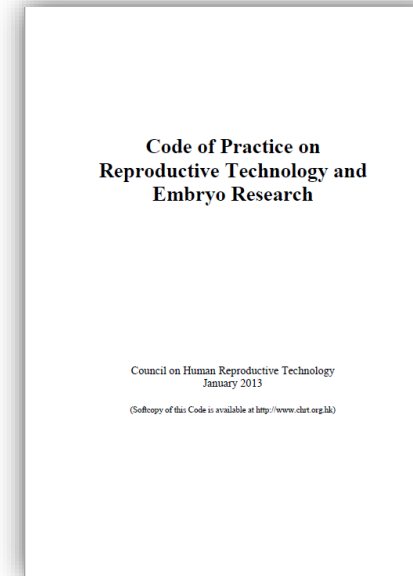
- Help couples **choose** the type of screening or diagnostic test that best suits their need
- Help couples **understand** the impact of fetal condition on the family
- Help couples choose the course of action for management of fetal abnormality that they will **not regret**





Ethical principles

- Autonomy (informed consent, non-directive counselling, full disclosure, voluntary)
- Nonmaleficence
- Beneficence
- Justice (indicated, accessible)



HKCOG Guidelines on Ethical Issues

Number 1

November 2002

Guidelines on Termination of Pregnancy in relation to Congenital Anomalies

published by The Hong Kong College of Obstetricians and Gynaecologists

A Foundation College of Hong Kong Academy of Medicine



- with severe anomaly
- may be offered as an option
- refer for second opinion if doctor has conscientious objection to pregnancy termination



Conclusion

- Aware of maternal and fetal reproductive risks
- Risks identification
- Options for at-risk couples
- Counselling concerns

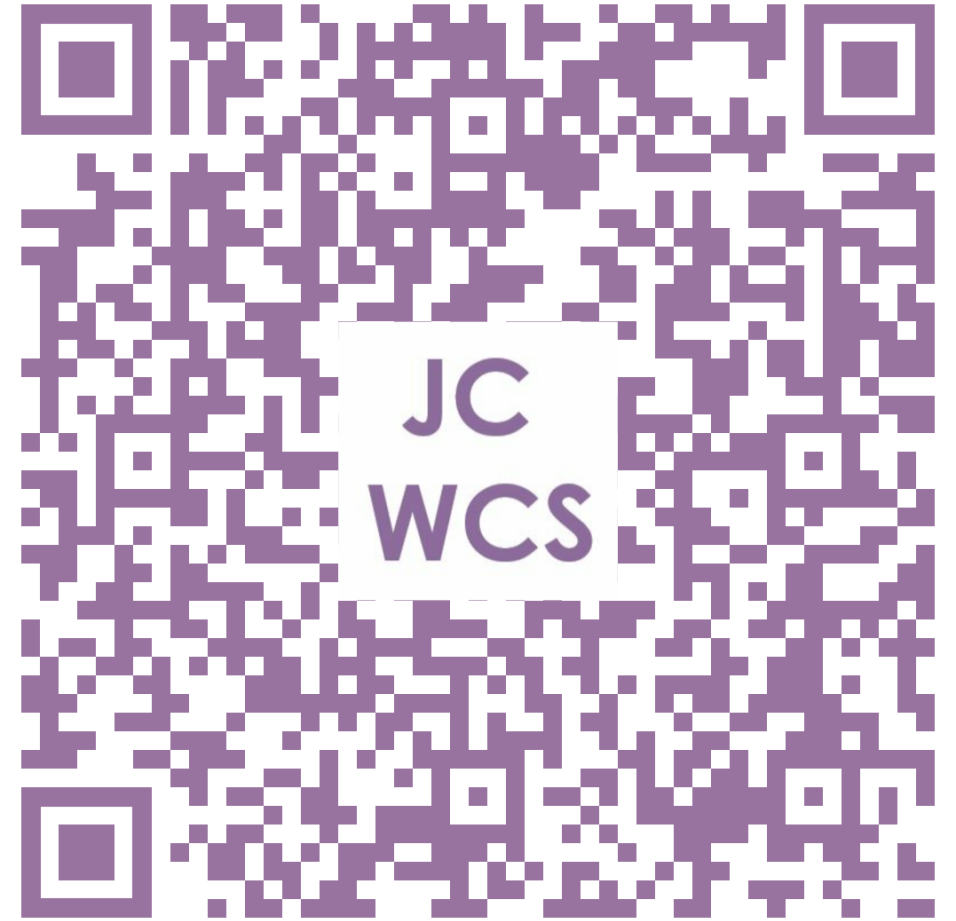


TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)